

ZEWEI DENG

Ph.D. Student in Industrial and Systems Engineering | University of Minnesota Twin Cities

deng0379@umn.edu | Website | Google Scholar | GitHub

U.S. Work Authorization · No Employer Sponsorship or CPT Required

PROFILE

Develop statistical, coding-theoretic, and optimization methods for trustworthy AI, with research on LLM watermarking and provenance, distributionally robust inference, uncertainty-aware sequential modeling, and temporal point processes.

EDUCATION

University of Minnesota Twin Cities

Ph.D. in Industrial and Systems Engineering; Advisor: Prof. Liyan Xie

2025–Present
Minneapolis, MN

University of Chicago

M.S. in Applied Data Science

2023–2024
Chicago, IL

University of Illinois Urbana-Champaign

B.S. in Statistics; Minors in Computer Science and Economics; GPA: 3.9/4.0

2019–2023
Urbana-Champaign, IL

RESEARCH & PROFESSIONAL EXPERIENCE

Graduate Research Assistant, University of Minnesota Twin Cities

Advisor: Prof. Liyan Xie

Summer 2026–Present
Minneapolis, MN

- Develop statistical and coding-theoretic methods for LLM watermark provenance, edit detection, local integrity verification, and watermarkability.
- Led an ECC watermark project accepted to EMNLP 2026 Findings, spanning method design, implementation, experiments, and manuscript preparation.
- Develop uncertainty-aware sequential models and distributionally robust inference for delayed point processes under reporting delays and model misspecification.

Research Assistant, Chicago Booth School of Business

Advisor: Prof. Dacheng Xiu

2024
Chicago, IL

- Built financial-text forecasting pipelines using BERT, LLaMA, and T5 representations with regression models for return and debt-related prediction.

Machine Learning Capstone, Alliant Credit Union / UChicago DSI

Client-facing machine-learning project

2024
Chicago, IL

- Contributed to model development and evaluation for a credit-union analytics application.

PUBLICATIONS & WORKING PAPERS

Detecting Post-generation Edits to Watermarked LLM Outputs via Error-Correcting Codes.

Zewei Deng, Muhammad Siddeek, Liyan Xie, Mohamed Seif, Mengdi Wang, H. Vincent Poor, and Andrea Goldsmith.
EMNLP Findings, 2026.

Local Integrity Checking for Watermarked LLM Outputs via Error-Correcting Codes.

Zewei Deng, Muhammad Siddeek, Liyan Xie, Mohamed Seif, Mengdi Wang, H. Vincent Poor, and Andrea Goldsmith.
Preprint, 2026; planned submission to Transactions on Machine Learning Research (TMLR).

Distributionally Robust Inference for Delayed Point Processes.

Zewei Deng, Shuyi Chen, Shixiang Zhu, and Liyan Xie.

Working paper in preparation for submission to the Annals of Applied Statistics (AOAS); workshop version forthcoming at the INFORMS Data Mining and Decision Analytics Workshop, 2026.

Watermarkability: Measuring and Improving Language-Model Compatibility with Logit-Based Watermarks.

Zewei Deng, Mohamed Seif, and Liyan Xie.

Preprint, 2026.

HawkesLLM: Semantic Uncertainty Propagation in Agentic Text Simulation.

Zewei Deng, Tinghan Ye, and Liyan Xie.

ICML Workshop on Statistical Frameworks for Uncertainty in Agentic Systems, 2026. [Paper]

Learning to Test: Physics-Informed Representation for Dynamical Instability Detection.

Minxing Zheng, **Zewei Deng**, Liyan Xie, and Shixiang Zhu.

arXiv:2604.10967, 2026. [Paper]

TEACHING & HONORS

Teaching: Teaching Assistant, University of Minnesota Twin Cities, Sep 2025–May 2026

Honors: Highest Distinction in Statistics; Dean's List (2020–2023); COMAP MCM Honorable Mention (2021)